Part-1: Ordinary Differential equation.

Write down the degree and order of the following DE:

(1) $\frac{d}{dx}\left(\frac{d^3y}{dx^3}\right) + \left(\frac{d^3y}{dx^3}\right)^3 + \frac{dy}{dx} + y = 0$

(2) $\left(\frac{\partial^3 u}{\partial x^3}\right)^2 + \frac{\partial^3 u}{\partial y^2} + \frac{\partial u}{\partial x} = 0$

(3) $\sqrt{\frac{dy}{dx} + y} = \frac{dy}{dx}$

(4) $\frac{d^4y}{dx^4} + \left(\frac{d^3y}{dx^3}\right)^2 + y = x$

(5) Find the differential equations of all circles in the XOY plane having centres on the x-axis and have given radius.

(6) Find the 2nd order differential equation of all circles having centres on the x-axis and have given radius.

(7) Find the 1st order differential equation of all circles having centres on the x-axis and passing through the origin.

(8) Find the differential equation corresponding to the family of curves $y = k(x-k)^2$, where k is an arbitary constant.

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(9) Find the differential equation of parabolas having latus rectum $4a$ and axis parallel to the y-axis.

Seperation of variable:

(10) Solve the DE $(x^2 - yx^2)\frac{dy}{dx} + (y^2 + xy^2) = 0$

(11) Identify and solve the DE

(i) $xy\frac{dy}{dx} = \frac{1+y^2}{1+x^2}(1+x+x^2)$

(ii) $\sqrt{1+x^2}\sqrt{1+y^2}\,dx + xy\,dy = 0$

(iii) $\frac{dy}{dx} = \frac{x+y-1}{x+y+1}$

Homogeneous DE:

(12) Identify and solve the DE:

(i) $(x^2 + xy)\,dy = (x^2 + y^2)\,dx$

(ii) $\frac{dy}{dx} = \frac{y^3 + 3x^2y}{x^3 + 3xy^2}$

(iii) $x^2y\,dx - (x^3 + y^3)\,dy = 0$

(iv) $\{x\sqrt{x^2+y^2} - y^2\}\,dx + xy\,dy = 0$

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Non-homogeneous DE:

(13) Identify and solve

(i) $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$

(ii) $\frac{dy}{dx} = \frac{3y-x+7}{x-7y-3}$

(iii) $\frac{dy}{dx} = \frac{3y-7x-7}{3x-7y-3}$

Linear DE:

(14) Identify and solve (linear ODE)

(i) $(1+x^2)\frac{dy}{dx} + 2xy - 4x^2 = 0$

(ii) $\frac{dy}{dx} + y\tan x = x^2\cos^3 x$

(iii) $\frac{dy}{dx} + y\sec x = \tan x$

Non linear DE:

(15) Identify and solve the DE

(i) $x\frac{dy}{dx} + y = y^2\ln x$

(ii) $\frac{dy}{dx} = x^3y^3 - xy$

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Exact DE:

(16) Identify and solve

(i) $(x^2+y^2)\,dx + 2xy\,dy = 0$

(ii) $(x^2-2xy-y^2)\,dx - (x+y)^2\,dy = 0$

(iii) $(x+2y-3)\,dy - (2x-y+1)\,dx = 0$

(iv) $(x^3+3xy^2)\,dx + (3x^2y+y^3)\,dy = 0$

Non exact DE:

(17) Identify and solve.

(i) $(x^2y-2xy^2)\,dx - (x^3-3x^2y)\,dy = 0$.

(ii) $(xy\sin xy+\cos xy)\,y\,dx + (xy\sin xy-\cos xy)\,x\,dy = 0$

(iii) $x^2y\,dx - (x^3+y^3)\,dy = 0$

(iv) $(x^2y^2+xy+1)\,y\,dx + (x^2y^2-xy+1)\,x\,dy = 0$

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Non-exact DE :

1. Identify and solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$
2. Solve $(xy\sin xy + \cos xy)y\,dx + (xy\sin xy - \cos xy)x\,dy = 0$

H.W

1. $x^2y\,dx - (x^3 + y^3)dy = 0$
2. $(x^2y^2 + xy + 1)y\,dx + (x^2y^2 - xy + 1)x\,dy = 0$

Higher Degree, First Order ODE:

1. Equation solvable for y: $y + px = x^4p^2$
2. Equation solvable for x: $y = 2px + y^2p^3$

H.W

Find the general solution and singular solution from

1. $y = 2px + p^4x^2$
2. $y = 2px - p^2$

Higher Order Linear differential equation with constant coefficients :

1. Solve the DE $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{2x}$

H.W

Find the complementary function and particular integral of

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1. $(D^2-3D+2)y = e^{5x}$

2. $(D^2+4D+4)y = e^{3x} - e^{-2x}$

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3. Find the general solution of $(D^2-2D+1)y = \cos 3x$

3. Find the complementary function and particular integral of $(D^3-D^2-6D)y = x^2+1$

4. Solve $(D^2+3D+2)y = e^{2x}\sin x$

H.W

1. Find the general solution of $(D^2-9)y = x+e^{2x}-\sin 2x$

2. Find the general solution of $(D^2-4D+4)y = x^3e^{2x}$

Application of ODE:

1. The rate at which the ice melts is proportione to the amount of ice at that instant. Find the amount of ice left after 2 hours if half of the quantity melts in 30 minutes.

2. Under certain conditions, cane sugar is converte into dextrose at a rate, which is proportiona to the amount unconverted at any time. If out of 75g of cane sugar at $t=0$, 8gms are converted during first 3 minutes, find the amount of cane sugar converted in 90 minutes.

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CSF-18th

MATH-1241

Partial Differential Equation

(1) Eliminating arbitary Constants:

i. form a partial differential equation by eliminating h and k from the equation.

$$(x-h)^2+(y-k)^2+z^2=\lambda^2$$

Homework:

Example:06 $\quad axe^{y}+\left(\frac{1}{2}\right)a^2e^{2y}+b$

Example:09

From partial differential equation by eliminating constant A and p from

$$z=Ae^{pt}\sin px.$$

Example:11

Eliminate a,b,c from $z=a(x+y)+b(x-y)+abt+c$

CT:04 (9:02)

Find the partial differential equation of all spheres of radius a, having centre on the yz plane.

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Eliminating Arbitrary function:

(i) $z = f(x^2 - y^2)$

(ii) $z = x + y + f(xy)$

(iii) $z = xy + f(x^2 + y^2)$

Lagrange's Method:

Type (I):

Solve the PDE $p\tan x + q\tan y = \tan z$ by lagrange's method.

Type (II):

(i) $p + 3q = 5z + \tan(y - 3x)$ solve this eqn by lagrange's method.

(ii) $xyp + y^2q = zxy - 2x^2$

(iii) $yzp + zxq = xy$

Type (III):

(i) $z(x+y)p + z(x-y)q = (x^2 + y^2)$

(ii) $(xy - zx)p + (yz - xy)q = zx - zy$

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CT:04 (9:03)

Which method do you prefer to solve a linear partial differential equation of order one? Using your chosen method Solve the partial DE equation $y^2p - xyq = x(z-2y)$

Charpits Method

Q1: Solve the PDE $px + qy = pq$ by charpits method.

Q2: Solve the PDE $zpq = p+q$ by charpits method.

Q3: Solve the PDE $q = 3p^2$ by charpits method.

Higher order linear PDE with constant co-efficient

Q1. Write down the working principle of higher order linear PDE.

Q2: Solve $(D^2 + DD' - 2D'^2)z = e^{2x+3y}$

Q3: Solve the PDE $(D^2 - 2DD' + D'^2)z = x+2y$

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Q4: $(D^2 - 5DD' + 4D'^2)z = \sqrt{3x+2y} + e^{2x+3y}$

Q5: $(D^3 - 4D^2D' + 4DD'^2)z = 6\sin(2x+3y) + e^{-2x+y}$

Q6: $(D^3 - 7DD'^2 + 6D'^3)z = \cos(x+2y) + \sqrt[5]{x+3y}$

OR,

$$\frac{\partial^3 z}{\partial x^3} - 7 \cdot \frac{\partial z}{\partial x} \cdot \frac{\partial^2 z}{\partial y^2} + 6 \cdot \frac{\partial^3 z}{\partial y^3} = \cos(x+2y) + \sqrt[5]{x+3y}$$

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Co-ordinate Geometry

1. Reduce the equation $32x^2+52xy-7y^2-64x-52y-148=0$

H.W

2. Test the nature of the conic $x^2+12xy-4y^2-6x-4y+9=0$ and find the centre of the conic. Also reduce the equation to the standard form.

Change of axes

1. Determine the equation of the curve $2x^2+3y^2-8x+6y-7=0$ when origin is transffered to $(2,-1)$

2. Determine the equation of the curve $x^2-2xy+y^2-2x-4y+3=0$ after rotating the axes through an angle $45°$.

3. Transform the equation $11x^2+24xy+4y^2-20x-40y-5=0$ to rectangular axes through the point $(2,-1)$ and inclined at an angle $\tan^{-1}4/3$

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- Conditions of co-ordinate geometry / general equation of second Degree
- co-ordinate system:
- change of axes

1. change of origin but no rotation of axes

2. Rotation of axes but origin fixed

3. Change of origin with rotation of the axes.

- Standard equation of conic section

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